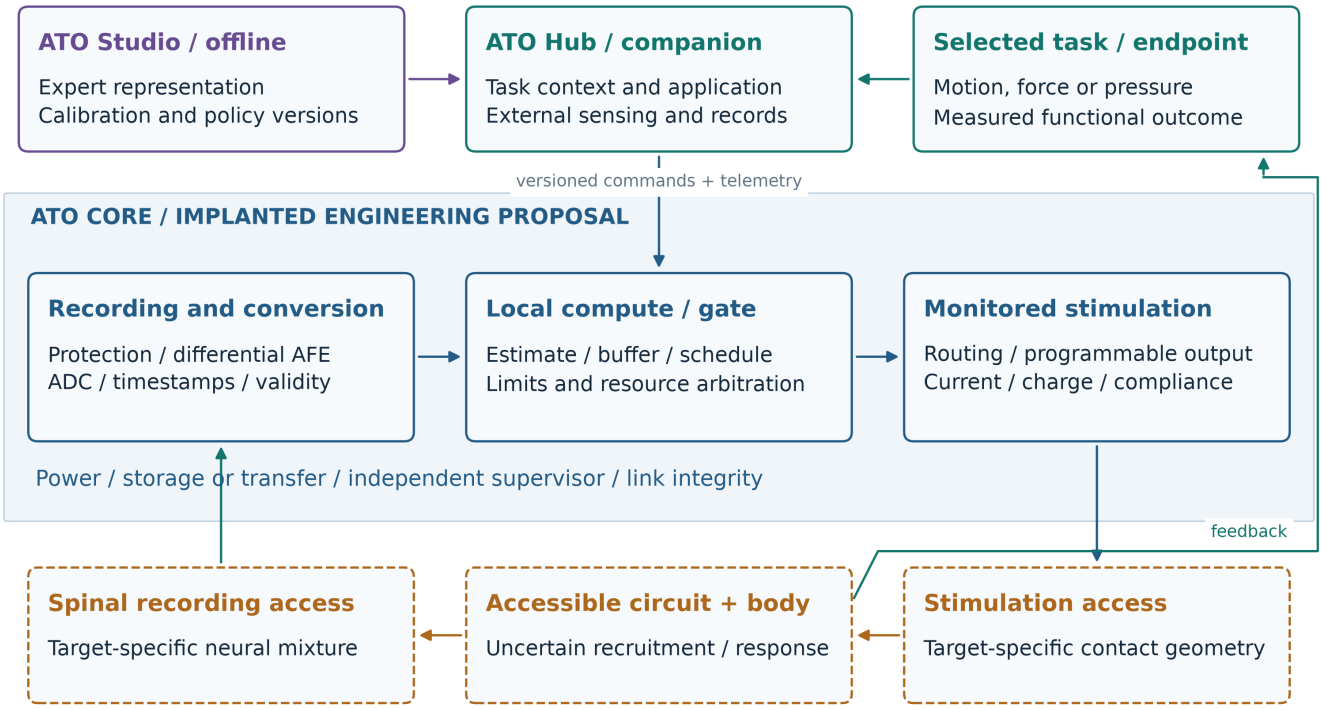
A spinal interface with a complete system concept

Mechanism and requirements dossier

An entirely digital proposal for invasive spinal recording, computational interpretation, personalized stimulation and feedback. The two ambitions are retained motor learning informed by expert recordings and automation of selected physiological responses.



Solid blue blocks are proposed engineering responsibilities; dashed amber blocks contain task-specific biological access or response assumptions. The drawing establishes a system boundary, not an implemented implant.

Design status: mechanism and requirements for review. Numerical budgets, native electronics, embedded implementation and implant/companion CAD follow in later stages.

AstroSam's Mind-Link inspires the progression from future vision to inspectable design. The SCCI application, mechanism, uncertainty analysis and engineering work are ATO's contribution.

What the proposed device would do

ATO would collect target-specific spinal signals, combine them with useful external measurements, estimate a limited task or bodily state, select a personalized intervention and monitor the delivered electrical output and functional response.

Part	Responsibility
ATO Core / implant	Acquire and digitize neural voltages; synchronize and buffer data; admit bounded commands; schedule and route programmable outputs; monitor current, charge and compliance; enforce supervision; manage power and telemetry.
ATO Hub / companion	Provide task context and external measurements; run session management and budgeted application services; manage records and operator interaction.
ATO Studio / offline	Process expert recordings, learn task representations, analyze learner calibration, evaluate models and version the resulting policies.
Target configuration	Supply recording reference/montage and stimulation geometry for the selected anatomical pathway. Cervical, lumbosacral and thoracic variants differ.

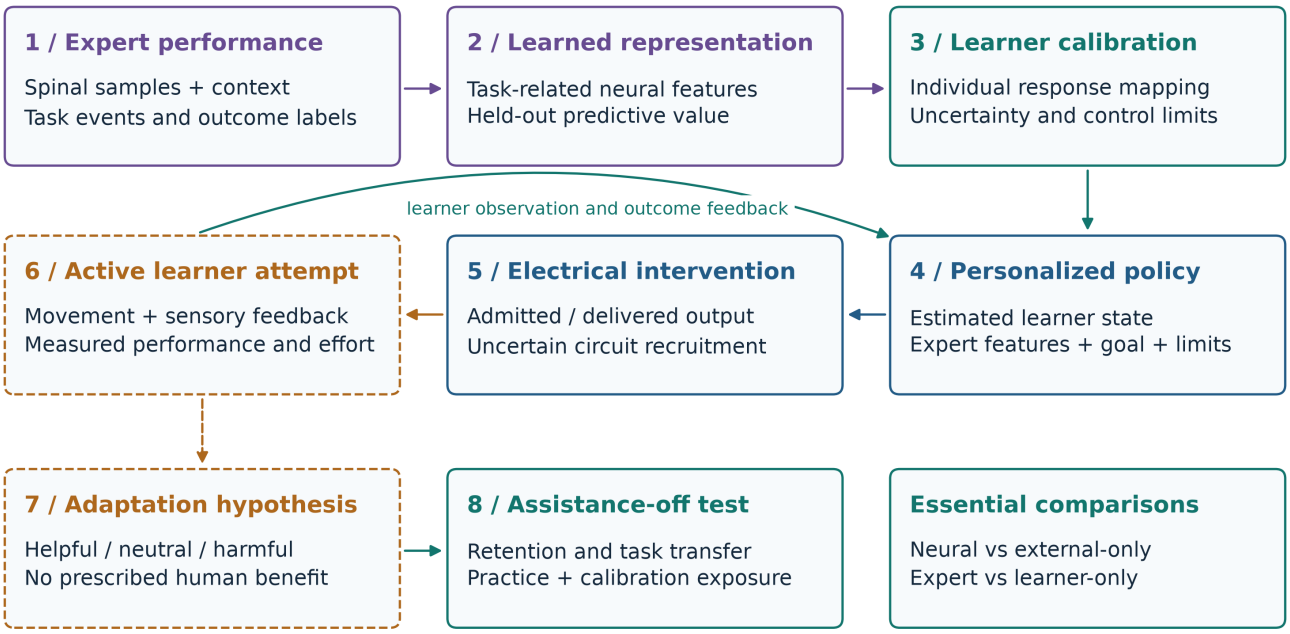
Three kinds of claim

Published component evidence describes what a particular study demonstrated. **Biological hypotheses** describe what must become possible for the consumer vision. **Digital verification** can establish performance of specified models and engineering designs within declared envelopes. None substitutes for the others.

The old v1.0.0 circuit and desktop enclosure remain subsystem references. New electrode counts, implant geometry and stimulation parameters are not inherited from those artifacts.

Full-system design is still in progress. This stage reports no new efficacy experiment, fabricated hardware or tested implantation. Website and video work remain parked.

Expert recordings become a personalized training policy



Transfer object: task-relevant information. Retention is a separate biological hypothesis.

The expert contributes a learned, task-related representation of spinal recordings with context and outcomes. The learner contributes a different body and neural system; a calibration identifies which available interventions influence the chosen task component.

The controller receives measured samples and estimated state with uncertainty. It proposes a bounded electrical intervention during active practice. It receives no hidden true skill or future trajectory. Literal voltage replay is not a functional inverse.

Separate learning link: appropriate assistance could improve practice or pairing, but it could also impair sensory feedback, reduce effort or create dependence. Helpful, neutral and harmful adaptation must all be allowed. Retention is evaluated after assistance is removed; online improvement alone is insufficient.

The starting paper's H1 is preserved. Spinal plasticity and rehabilitation provide component support [4, 5], while healthy expert-to-novice transfer remains unestablished.

Make the two information claims testable

Policy family	External measurements only	Neural + external measurements
Learner-only adaptation	Personalized learner baseline	Tests added neural information
Expert-informed adaptation	Tests expert task/context information	Full proposed ATO mechanism

The crossed comparison separates expert information from spinal information. It prevents a favorable expert trajectory or extra calibration exposure from being mistaken for an advantage of expert neural recordings.

Outcome	Required interpretation
Assistance-on	How much task performance changes while intervention is active.
Assistance-off retention	What performance remains after a predeclared removal interval.
Transfer	Whether retained improvement generalizes to an untrained related condition.
Acquisition exposure	Practice required to meet and retain a predefined criterion; include calibration cost and trials that never reach the criterion.

Challenge the mechanism

Use held-out learners and contexts, shuffled expert timing, incorrect learner mappings, ambiguous observations, stimulation artifacts, delay, saturation, dropout and insufficient authority. Keep an oracle comparison explicitly labelled. A null or harmful result is a valid and useful result of the digital study.

A 1.5x or 2x learning gain is an ambition, not a supplied parameter or measured fact. Motor execution is only part of piano or football expertise; cognitive and perceptual components remain distributed.

Physiological automation follows a named endpoint



Feedback follows the functional outcome, with freshness and adverse-effect checks.

Reference: pressure regulation after SCI

External pressure / lower-thoracic pressor access
No assumed universal ability to lower pressure

Application-specific conclusion

A triggered event does not establish its postponement
Recording must earn its added information value

A regulator needs a useful measured variable and bounded restoring influence. Triggering a response needs causal access to the action. Repetition additionally needs recovery and outcome feedback. Maintaining a state or delaying its termination may require selective opposing access that triggering evidence does not establish.

Blood-pressure regulation after SCI is a strong initial reference [6]. Its feedback remains external pressure measurement, principally systolic pressure in that precedent. A spinal readout may provide recruitment information, but its additional value must be tested. A stimulation-plus-external-feedback mode is accurately described as such within the wider recording/stimulation platform.

Bladder storage, voiding, cough, respiratory support and selected peripheral sexual responses require different endpoints and configurations. No universal body-control or subjective-experience claim follows from this architecture.

Choose references without shrinking the future platform

Candidate	Useful starting point	Unresolved leap
Lower-limb coordination	Locomotor recruitment and timing studies; footwork motivates a consumer coordination task.	Healthy retained expert skill transfer; neural value beyond motion/EMG; dynamic balance.
Cervical / piano coordination	Upper-limb stimulation studies support access to movement-related circuits.	Independent finger readout/control, task representation and useful retention without sensory interference.
Pressure regulation	Human lower-thoracic stimulation with pressure feedback gives a defined pathway and controlled variable.	Healthy use, optional neural recording value, bounded pressor authority and side effects.

Provisional recommendation: lower-limb coordination and pressure regulation as the first quantitative reference configurations. Keep the cervical/piano and other physiological variants in the capability map. These are reference cases for platform engineering, not evidence of consumer efficacy.

Access candidate	Tradeoff to resolve
Shared epidural recording/stimulation	Simpler shared access; evoked responses are distinct from decoding spontaneous expert activity.
Intradural surface recording + epidural stimulation	Comparison baseline with separate signal/recruitment access; artifacts, chronic stability and human integration remain unresolved.
Penetrating recording with selected stimulation access	Potential local access; sampling bias, tissue response, durability and mechanical burden increase.

Surface and intraspinal recording studies [1, 2] motivate alternatives, not a selected chronic human decoder. Counts and placements follow signal/control requirements. Spinal segment, root, vertebral level and actual electrode position remain distinct coordinates.

Engineering budgets must close before detailed CAD

The detailed dossier contains 36 requirements, 14 explicit assumptions and 14 capability entries. The tables define responsibilities and staged checks. This brief summarizes the quantities that Stage 2 must make numerical.

Budget	Quantities and constraints
Acquisition	Signal and electrode offset ranges; input noise; bandwidth; ADC resolution; montage; artifact saturation and recovery.
Data and compute	Sampled channels x sample rate x bits, framing, timestamps, buffers, estimation cost and memory; contact count is a separate quantity.
Timing	Filtering + conversion + recovery + computation + transport + scheduling; freshness, jitter and clock error in each operating mode.
Stimulation	Routing and independent sources; load/polarization voltage plus driver headroom; actual integrated current per phase; charge and duty-cycle limits.
Power and heat	Acquisition, compute, telemetry, monitoring, storage/transfer and stimulation energy; realistic modes, efficiency and thermal uncertainty.
Mechanical interfaces	Target geometry; conductors and feedthroughs; populated board, energy-storage, cable and connector envelopes; stiffness and fit assumptions.

Fault behavior is application-specific

Radio loss, stale measurements, brownout, reset and interrupted phases need independent handling. Disable new output admission on invalid state; track delivered charge and remaining power. An interrupted pulse does not justify an automatic opposite pulse. Removing assistance can also carry physiological risk, so fallback behavior requires target-specific review.

Operating modes include initialization, self-test, recording, expert collection, learner calibration, assisted practice, unassisted assessment, physiological regulation, charging and fault recovery. Concurrent applications need explicit resource arbitration.

What verification can establish

Work package	Completion evidence	Remaining boundary
Stage 1 / specification	Coherent mechanism chains, responsibilities, source audit, requirements, assumptions, links and reviewed figures.	A proposal has not been built.
Stage 2 / numerical architecture	Closed signal, timing, output, energy and mechanical envelopes with uncertainty.	Budgets depend on declared load and access models.
Stage 3 / computational mechanism	Observable-only model, fair comparisons, held-out calibration and negative results.	Model adaptation is not human learning.
Stages 4-5 / electronics and CAD	Native designs, monitored waveform models, fault tests, matching pinouts and interference/export checks.	No fabricated hardware or implantation validation.
Stage 6 / integrated research release	End-to-end digital trace, reproducibility, source-linked manuscript and repository review.	Biological efficacy, chronic stability and clinical use need separate future evidence.

Recommended next decision

Review the motor reference and physiological endpoint before fixing target anatomy. Then calculate channel scaling, waveform/load and timing budgets; compare access options against those requirements. If information or control authority is insufficient, revise the capability or configuration before board routing and implant geometry.

Full source and design records

docs/stage1/README.md is the entry point. The source audit records study-specific support and limits; the requirements and verification matrix map future checks. The Stage 1 verification receipt records only checks actually performed. The original 28-page source is preserved read-only with its stated 30 September 2026 cutoff.

Selected primary foundations

[1] El Hadwe et al. (2026). Surface circumferential spinal cord recording in freely moving rodents. Nature Communications.
https://doi.org/10.1038/s41467-026-75128-z
Spinal surface observations in animal studies; not a chronic human expert decoder.

[2] Fan et al. (2023). A Hyperflexible Electrode Array for Long-Term Recording and Decoding of Intraspinal Neuronal Activity. Advanced Science.
https://doi.org/10.1002/advs.202303377
Alternative recording access and mouse decoding; human integration remains separate.

[3] Formento et al. (2018). Electrical spinal cord stimulation must preserve proprioception to enable locomotion in humans with spinal cord injury. Nature Neuroscience.
https://doi.org/10.1038/s41593-018-0262-6
Sensory interference matters when choosing timing and recruitment.

[4] Nishimura et al. (2013). Spike-timing-dependent plasticity in primate corticospinal connections induced during free behavior. Neuron.
https://doi.org/10.1016/j.neuron.2013.08.028
Hybrid cortical-triggered spinal plasticity; not spinal-only expert transfer.

[5] McPherson et al. (2015). Targeted, activity-dependent spinal stimulation produces long-lasting motor recovery in chronic cervical spinal cord injury. PNAS.
https://doi.org/10.1073/pnas.1505383112
Activity-linked rehabilitation in injured rats; not a healthy learning-speed multiplier.

[6] Phillips et al. (2025). An implantable system to restore hemodynamic stability after spinal cord injury. Nature Medicine.
https://doi.org/10.1038/s41591-025-03614-w
Human target-specific pressure feedback and lower-thoracic stimulation; external pressure measurement.

[7] de Freitas et al. (2026). Spinal cord stimulation for upper limb motor function in people with chronic post-stroke hemiparesis: a feasibility trial. Nature Medicine.
https://doi.org/10.1038/s41591-026-04435-1
Upper-limb access component evidence; not independent healthy piano expertise transfer.

Evidence handling

This is a narrative design audit, not a systematic review. Study counts, access scope and important extrapolation limits are in docs/stage1/source-audit.csv. Use those records for study endpoints, access scope and unverified details.

Control foundations: Astrom and Murray, Feedback Systems, author-hosted book. Author-hosted text. Observability and bounded control authority constrain the interface even if computation becomes ideal.

Inspiration: AstroSam / Mind-Link repository and the video identified in the project introduction. Engineering presentation inspiration is distinct from SCCI evidence.